

WASP-10b

R-1466

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_141022 · created 2026-08-01T11:22:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456952.6606 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1772 +/- 0.0083
Transit depth (Rp/Rs)^2	3.14 +/- 0.29 [%]
Orbital Inclination (inc)	88.3 +/- 0.71
Airmass coefficient 1 (a1)	0.9753 +/- 0.0097
Airmass coefficient 2 (a2)	0.026 +/- 0.021
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	#3 - [177, 231]
Optimal Aperture	3.67
Optimal Annulus	10.98
Transit Duration (day)	0.0921 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1772 $\pm$ 0.0083 vs 0.1592 (deviation 2.17 σ)
Duration fit vs published	0.0921 d vs 0.0946 d (deviation 0.65 σ)
Mid-time O-C	5.3 min
Depth SNR	21.3
Residual scatter	1.0 %

Light curve

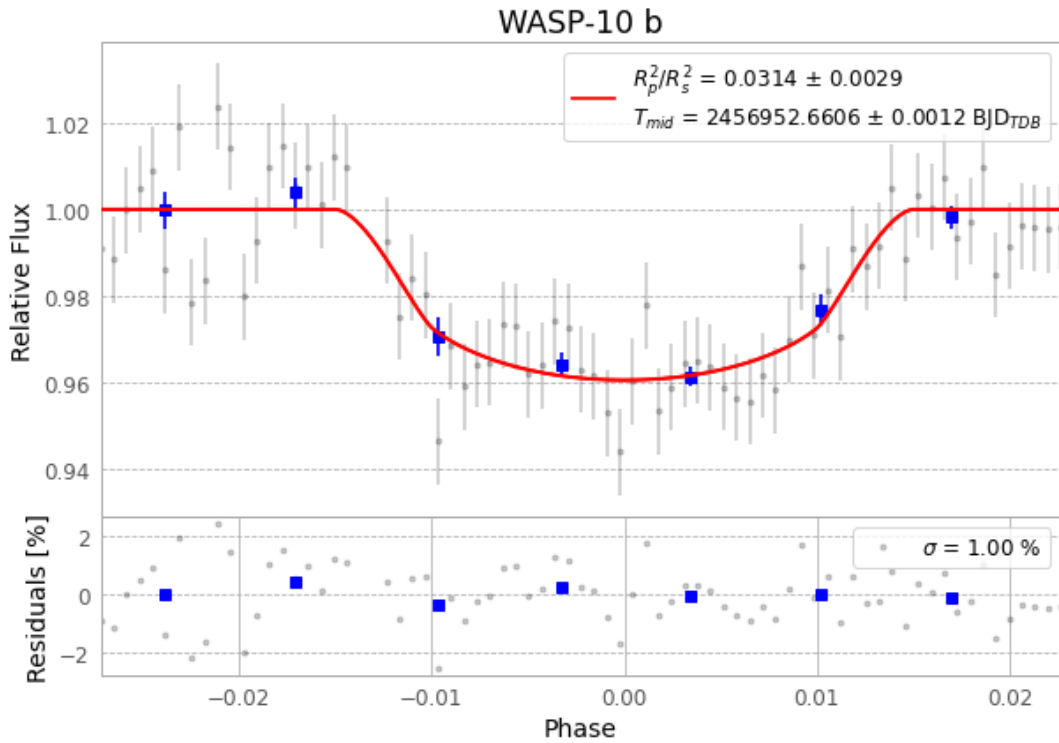


Figure 1 — FinalLightCurve_WASP-10 b_2014-10-21.png (SHA-256 c438b7f5f6a3c88d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [235, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 88f714122a30b71d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-10141022014216.FITS → WASP-10141022052412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-141022.log (74b2e91808e2...), FinalLightCurve_WASP-10 b_2014-10-21.png (c438b7f5f6a3...), FinalLightCurve_WASP-10 b_2014-10-21.csv (9441f37191bd...)

Compute resources

Wall time 220.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 11:22Z.